



III. MATRICE

III. 4. Regularnost i inverz matrice

Kažemo da je matrica $A \in M_n$ (kvadratna matrica) **regularna** ako postoji matrica $B \in M_n$ takva da vrijedi

$$AB = BA = I.$$

U tom slučaju matrica B zove se **multiplikativni inverz** ili **inverzna** matrica od A i označava s A^{-1} .

Za matricu $A \in M_n$ koja nema multiplikativni inverz kažemo da je **singularna**.

Matrica $A \in M_n$ je regularna ako i samo ako je $\det A \neq 0$, u suprotnom je matrica singularna, tj.

*Matrica kojoj je determinanta jednaka nuli naziva se **singularna matricama**.*

Primjer 1. Matrica B u zadatku 10. pod b) je singularna jer smo dobili da je njena determinanta jednaka nuli.

Svojstva inverznih matrica:

- Singularna matrica nema inverz.
- Inverz, ako postoji, jednoznačno je određen.
- $\det A^{-1} = \frac{1}{\det A}$.
- $(AB)^{-1} = B^{-1}A^{-1}$.

1.1. Određivanje inverza pomoću matrice kofaktora

$$A^{-1} = \frac{1}{\det A} (A^*)^T, \quad A^* - \text{matrica kofaktora}$$

Zadatak 1. [Formula za inverz matrice 2x2]: Odredite inverz matrice tipa 2x2 zadane s

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Rješenje:

$$A^{-1} = \frac{1}{\det A} (A^*)^T$$

$$\det A = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

$$A_{ij} = (-1)^{i+j} \cdot \det M_{ij} \text{ pa je :}$$

$$A_{11} = (-1)^{1+1} \cdot d = d \quad (\text{uklonili prvi redak i prvi stupac})$$

$$A_{12} = (-1)^{1+2} \cdot c = -c \quad (\text{uklonili prvi redak i drugi stupac})$$

$$A_{21} = (-1)^{2+1} \cdot b = -b \quad (\text{uklonili drugi redak i prvi stupac})$$

$$A_{22} = (-1)^{2+2} \cdot a = a \quad (\text{uklonili drugi redak i drugi stupac})$$



$$A^* = \begin{bmatrix} d & -c \\ -b & a \end{bmatrix}$$

Inverz matrice A je:

$$A^{-1} = \frac{1}{\det A} (A^*)^T = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Zadatak 2. Odredite inverz sljedećih matrica

a) $A = \begin{bmatrix} 2 & -4 \\ -3 & 1 \end{bmatrix}$

Rješenje: Odredimo najprije determinantu matrice A :

$$\det A = \begin{vmatrix} 2 & -4 \\ -3 & 1 \end{vmatrix} = 2 \cdot 1 - (-3) \cdot (-4) = 2 - 12 = -10.$$

$$A_{11} = (-1)^{1+1} \cdot 1 = 1 \quad (\text{uklonili prvi redak i prvi stupac})$$

$$A_{12} = (-1)^{1+2} \cdot (-3) = 3 \quad (\text{uklonili prvi redak i drugi stupac})$$

$$A_{21} = (-1)^{2+1} \cdot (-4) = 4 \quad (\text{uklonili drugi redak i prvi stupac})$$

$$A_{22} = (-1)^{2+2} \cdot 2 = 2 \quad (\text{uklonili drugi redak i drugi stupac})$$

$$A^* = \begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix}$$

Inverz matrice A je:

$$\Rightarrow A^{-1} = \frac{1}{\det A} (A^*)^T = -\frac{1}{10} \begin{bmatrix} 1 & 4 \\ 3 & 2 \end{bmatrix}.$$

b) $A = \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix}$

Rješenje:

- Odredimo najprije determinantu matrice A :

$$\begin{aligned} \det A &= \begin{vmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{vmatrix} = 3 \cdot \begin{vmatrix} 6 & -5 \\ -2 & 2 \end{vmatrix} - (-1) \cdot \begin{vmatrix} -15 & -5 \\ 5 & 2 \end{vmatrix} + 1 \cdot \begin{vmatrix} -15 & 6 \\ 5 & -2 \end{vmatrix} \\ &= 3(6 \cdot 2 - (-2) \cdot (-5)) + (-15 \cdot 2 - 5 \cdot (-5)) + (-15 \cdot (-2) - 5 \cdot 6) \\ &= 6 - 5 \\ &= 1 \end{aligned}$$



- Matrica kofaktora matrice A :

$$A_{11} = (-1)^{1+1} \cdot \begin{vmatrix} 6 & -5 \\ -2 & 2 \end{vmatrix} = 2$$

$$A_{21} = (-1)^{2+1} \cdot \begin{vmatrix} -1 & 1 \\ -2 & 2 \end{vmatrix} = 0$$

$$A_{31} = (-1)^{3+1} \cdot \begin{vmatrix} -1 & 1 \\ 6 & -5 \end{vmatrix} = -1$$

uklonili prvi redak i prvi stupac
matrice A

$$A_{12} = (-1)^{1+2} \cdot \begin{vmatrix} -15 & -5 \\ 5 & 2 \end{vmatrix} = 5$$

$$A_{22} = (-1)^{2+2} \cdot \begin{vmatrix} 3 & 1 \\ 5 & 2 \end{vmatrix} = 1$$

$$A_{32} = (-1)^{3+2} \cdot \begin{vmatrix} 3 & 1 \\ -15 & -5 \end{vmatrix} = 0$$

$$A_{13} = (-1)^{1+3} \cdot \begin{vmatrix} -15 & 6 \\ 5 & -2 \end{vmatrix} = 0$$

$$A_{23} = (-1)^{2+3} \cdot \begin{vmatrix} 3 & -1 \\ 5 & -2 \end{vmatrix} = 1$$

$$A_{33} = (-1)^{3+3} \cdot \begin{vmatrix} 3 & -1 \\ -15 & 6 \end{vmatrix} = 3$$

$$\rightarrow A^* = \begin{bmatrix} 2 & 5 & 0 \\ 0 & 1 & 1 \\ -1 & 0 & 3 \end{bmatrix}$$

$$\Rightarrow A^{-1} = \frac{1}{\det A} (A^*)^T = \begin{bmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix}$$

c) $A = \begin{bmatrix} 1 & -2 & 3 \\ 4 & 0 & 5 \\ -1 & 2 & 3 \end{bmatrix}$

Rješenje:

- Odredimo najprije determinantu matrice A :

$$\begin{aligned} \det A &= \begin{vmatrix} 1 & -2 & 3 \\ 4 & 0 & 5 \\ -1 & 2 & 3 \end{vmatrix} = 1 \cdot \begin{vmatrix} 0 & 5 \\ 2 & 3 \end{vmatrix} - (-2) \cdot \begin{vmatrix} 4 & 5 \\ -1 & 3 \end{vmatrix} + 3 \cdot \begin{vmatrix} 4 & 0 \\ -1 & 2 \end{vmatrix} \\ &= 1 \cdot (0 \cdot 3 - 5 \cdot 2) + 2(4 \cdot 3 - 5 \cdot (-1)) + 3(4 \cdot 2 - 0 \cdot (-1)) \\ &= -10 + 34 + 24 \\ &= 48 \end{aligned}$$

- Matrica kofaktora matrice A :

$$A_{11} = (-1)^{1+1} \cdot \begin{vmatrix} 0 & 5 \\ 2 & 3 \end{vmatrix} = -10$$

$$A_{21} = (-1)^{2+1} \cdot \begin{vmatrix} -2 & 3 \\ 2 & 3 \end{vmatrix} = 12$$

$$A_{31} = (-1)^{3+1} \cdot \begin{vmatrix} -2 & 3 \\ 0 & 5 \end{vmatrix} = -10$$

uklonili prvi redak i prvi stupac
matrice A

$$A_{12} = (-1)^{1+2} \cdot \begin{vmatrix} 4 & 5 \\ -1 & 3 \end{vmatrix} = -17$$

$$A_{22} = (-1)^{2+2} \cdot \begin{vmatrix} 1 & 3 \\ -1 & 3 \end{vmatrix} = 6$$

$$A_{32} = (-1)^{3+2} \cdot \begin{vmatrix} 1 & 3 \\ 4 & 5 \end{vmatrix} = 7$$

$$A_{13} = (-1)^{1+3} \cdot \begin{vmatrix} 4 & 0 \\ -1 & 2 \end{vmatrix} = 8$$

$$A_{23} = (-1)^{2+3} \cdot \begin{vmatrix} 1 & -2 \\ -1 & 2 \end{vmatrix} = 0$$

$$A_{33} = (-1)^{3+3} \cdot \begin{vmatrix} 1 & -2 \\ 4 & 0 \end{vmatrix} = 8$$

$$\rightarrow A^* = \begin{bmatrix} -10 & -17 & 8 \\ 12 & 6 & 0 \\ -10 & 7 & 8 \end{bmatrix}$$



$$\Rightarrow A^{-1} = \frac{1}{\det A} (A^*)^T = \frac{1}{48} \begin{bmatrix} -10 & 12 & -10 \\ -17 & 6 & 7 \\ 8 & 0 & 8 \end{bmatrix}.$$

1.2. Gauss – Jordanova metoda za određivanje inverza

Elementarne transformacije nad matricama zovemo sljedeće operacije:

1. Zamjena dva retka (dva stupca)
2. Množenje proizvoljnog retka (stupca) brojem različitim od nule
3. Množenje proizvoljnog retka (stupca) matrice brojem različitim od nule, i dodavanje bilo kojeg retku (stupcu) matrice

Definicija 1. Matrica $B \in M_{mn}$ je ekvivalentna matrici $A \in M_{mn}$ ako se B može dobiti iz A primjenom konačno mnogo elementarnih transformacija reda ili stupaca.)

- **Gaussov algoritam za određivanje inverzne matrice**

Inverznu matricu možemo odrediti pomoću elementarnih transformacija koje se vrše **samo nad redcima** (!!!)

$$[A|I] \sim \dots \sim [I|A^{-1}]$$

Odnosno, matrici $A \in M_n$ s desne strane nadodamo jediničnu matricu istog reda, tj. $I \in M_n$. Tako dobivamo novu matricu reda $(n, 2n)$. Zatim, provodimo elementarne transformacije nad redcima sve dok s lijeve strane ne dobijemo jediničnu matricu (ukoliko ne uspijemo dobiti jediničnu matricu – jedan redak je nul redak tada matrica **nema** inverz). Matrica s desne strane je tada inverz matrice A , tj. A^{-1} .

Sjetimo da se da je A^{-1} inverzna matrica ukoliko vrijedi: $A \cdot A^{-1} = A^{-1} \cdot A = I$.

Pogledajmo sljedeći primjer koji će nam zorno pokazati kako određujemo inverz matrice

Gaussovom metodom eliminacije.

Primjer 2. Odredite inverz matrice

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 7 & 8 \\ -3 & -1 & 12 \end{bmatrix}.$$

Rješenje:

$$[A|I] = \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 4 & 7 & 8 & 0 & 1 & 0 \\ -3 & -1 & 12 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\substack{\cdot(-4)/\cdot 3 \\ \swarrow + \quad \searrow \\ \swarrow +}} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & -1 & -4 & -4 & 1 & 0 \\ 0 & 5 & 21 & 3 & 0 & 1 \end{array} \right] \xleftarrow{\substack{\cdot 2 \quad \cdot 5 \\ \swarrow +}} \left[\begin{array}{ccc|ccc} 1 & 0 & -5 & -7 & 2 & 0 \\ 0 & -1 & -4 & -4 & 1 & 0 \\ 0 & 0 & 1 & -17 & 5 & 1 \end{array} \right] \xleftarrow{\substack{\cdot 4 \quad \cdot 5 \\ \uparrow}} \left[\begin{array}{ccc|ccc} 1 & 0 & -5 & -7 & 2 & 0 \\ 0 & -1 & -4 & -4 & 1 & 0 \\ 0 & 0 & 1 & -17 & 5 & 1 \end{array} \right] \xrightarrow{\substack{\cdot(-1) \\ \cdot(-4)/\cdot 3}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -92 & 27 & 5 \\ 0 & -1 & 0 & -72 & 21 & 4 \\ 0 & 0 & 1 & -17 & 5 & 1 \end{array} \right] \xrightarrow{\substack{\cdot(-1)}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -92 & 27 & 5 \\ 0 & 1 & 0 & 72 & -21 & -4 \\ 0 & 0 & 1 & -17 & 5 & 1 \end{array} \right] \xrightarrow{\substack{\text{A}^{-1}}}$$

Zadatak 3. Gauss – Jordanovom metodom odredite inverz sljedećih matrica:

Rješenje:

$$A^{-1} = \begin{bmatrix} -1/10 & -2/5 \\ -3/10 & -1/5 \end{bmatrix} = -\frac{1}{10} \begin{bmatrix} 1 & 4 \\ 3 & 2 \end{bmatrix}$$

Rješenje:

$$B^{-1} = \begin{bmatrix} 59 & -16 & 5 \\ 25 & -7 & 2 \\ 14 & -4 & 1 \end{bmatrix}$$

Rješenje:

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & -5 & 0 \\ 0 & 1 & 2 & 1 & -3 & 0 \\ 0 & 0 & 1 & 0 & 2 & 1 \end{array} \right] \leftarrow \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & -5 & 0 \\ 0 & 1 & 0 & 1 & -7 & -2 \\ 0 & 0 & 1 & 0 & 2 & 1 \end{array} \right] \underbrace{\hspace{1cm}}_{C^{-1}}$$



$$C^{-1} = \begin{bmatrix} 2 & -5 & 0 \\ 1 & -7 & -2 \\ 0 & 2 & 1 \end{bmatrix}$$

d) $D = \begin{bmatrix} 1 & -2 & 3 \\ 4 & 0 & 5 \\ -1 & 2 & 3 \end{bmatrix}$

Rješenje:

$$\begin{aligned} [D|I] &= \left[\begin{array}{ccc|ccc} 1 & -2 & 3 & 1 & 0 & 0 \\ 4 & 0 & 5 & 0 & 1 & 0 \\ -1 & 2 & 3 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\substack{R_2 \leftarrow R_2 - 4R_1 \\ R_3 \leftarrow R_3 + R_1}} \left[\begin{array}{ccc|ccc} 1 & -2 & 3 & 1 & 0 & 0 \\ 0 & 8 & -7 & -4 & 1 & 0 \\ 0 & 0 & 6 & 1 & 0 & 1 \end{array} \right] \xrightarrow{R_2 \leftarrow R_2 \cdot \frac{1}{8}} \left[\begin{array}{ccc|ccc} 1 & -2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -\frac{7}{8} & -\frac{1}{2} & \frac{1}{8} & 0 \\ 0 & 0 & 6 & 1 & 0 & 1 \end{array} \right] \xrightarrow{R_3 \leftarrow R_3 - 6R_2} \left[\begin{array}{ccc|ccc} 1 & -2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -\frac{7}{8} & -\frac{1}{2} & \frac{1}{8} & 0 \\ 0 & 0 & 6 & 1 & 0 & 1 \end{array} \right] \\ &\xrightarrow{R_3 \leftarrow R_3 \cdot \frac{1}{6}} \left[\begin{array}{ccc|ccc} 1 & -2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -\frac{7}{8} & -\frac{1}{2} & \frac{1}{8} & 0 \\ 0 & 0 & 1 & \frac{1}{6} & 0 & \frac{1}{6} \end{array} \right] \xrightarrow{R_1 \leftarrow R_1 + 2R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & -\frac{1}{4} & -\frac{1}{2} & \frac{1}{4} & 0 \\ 0 & 1 & -\frac{7}{8} & -\frac{1}{2} & \frac{1}{8} & 0 \\ 0 & 0 & 1 & \frac{1}{6} & 0 & \frac{1}{6} \end{array} \right] \xrightarrow{R_1 \leftarrow R_1 + \frac{1}{4}R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{5}{24} & \frac{1}{4} & -\frac{5}{24} \\ 0 & 1 & -\frac{7}{8} & -\frac{1}{2} & \frac{1}{8} & 0 \\ 0 & 0 & 1 & \frac{1}{6} & 0 & \frac{1}{6} \end{array} \right] \\ &\xrightarrow{R_2 \leftarrow R_2 + \frac{7}{8}R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{5}{24} & \frac{1}{4} & -\frac{5}{24} \\ 0 & 1 & 0 & -\frac{17}{48} & \frac{1}{8} & \frac{7}{48} \\ 0 & 0 & 1 & \frac{1}{6} & 0 & \frac{1}{6} \end{array} \right] \xrightarrow{R_2 \leftarrow R_2 \cdot 2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{5}{24} & \frac{1}{4} & -\frac{5}{24} \\ 0 & 1 & 0 & -\frac{17}{24} & \frac{1}{4} & \frac{7}{24} \\ 0 & 0 & 1 & \frac{1}{6} & 0 & \frac{1}{6} \end{array} \right] \\ &\Rightarrow D^{-1} = \begin{bmatrix} -\frac{5}{24} & \frac{1}{4} & -\frac{5}{24} \\ -\frac{17}{24} & \frac{1}{4} & \frac{7}{24} \\ \frac{1}{6} & 0 & \frac{1}{6} \end{bmatrix} = \frac{1}{48} \begin{bmatrix} -10 & 12 & -10 \\ -17 & 6 & 7 \\ 8 & 0 & 8 \end{bmatrix} \end{aligned}$$

e) $E = \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix}$

Rješenje:

$$\begin{aligned} [E|I] &= \left[\begin{array}{ccc|ccc} 3 & -1 & 1 & 1 & 0 & 0 \\ -15 & 6 & -5 & 0 & 1 & 0 \\ 5 & -2 & 2 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\substack{R_1 \leftarrow R_1 \cdot (-2) \\ R_2 \leftarrow R_2 + 5R_1 \\ R_3 \leftarrow R_3 - R_1}} \left[\begin{array}{ccc|ccc} 3 & -1 & 1 & 1 & 0 & 0 \\ -15 & 6 & -5 & 0 & 1 & 0 \\ -1 & 0 & 0 & -2 & 0 & 1 \end{array} \right] \xrightarrow{R_1 \leftarrow R_1 \cdot (-1)} \left[\begin{array}{ccc|ccc} 3 & -1 & 1 & 1 & 0 & 0 \\ -15 & 6 & -5 & 0 & 1 & 0 \\ 1 & 0 & 0 & 2 & 0 & -1 \end{array} \right] \\ &\xrightarrow{R_1 \leftarrow R_1 \cdot \frac{1}{3}} \left[\begin{array}{ccc|ccc} 1 & -\frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & -\frac{1}{3} \\ -15 & 6 & -5 & 0 & 1 & 0 \\ 1 & 0 & 0 & 2 & 0 & -1 \end{array} \right] \xrightarrow{R_2 \leftarrow R_2 + 15R_1} \left[\begin{array}{ccc|ccc} 1 & -\frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & -\frac{1}{3} \\ 0 & 2 & -6 & 5 & 1 & -5 \\ 1 & 0 & 0 & 2 & 0 & -1 \end{array} \right] \xrightarrow{R_2 \leftarrow R_2 \cdot \frac{1}{2}} \left[\begin{array}{ccc|ccc} 1 & -\frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & -\frac{1}{3} \\ 0 & 1 & -3 & \frac{5}{2} & \frac{1}{2} & -\frac{5}{2} \\ 1 & 0 & 0 & 2 & 0 & -1 \end{array} \right] \\ &\xrightarrow{R_1 \leftarrow R_1 + \frac{1}{3}R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & 0 & -1 \\ 0 & 1 & -3 & \frac{5}{2} & \frac{1}{2} & -\frac{5}{2} \\ 1 & 0 & 0 & 2 & 0 & -1 \end{array} \right] \xrightarrow{R_3 \leftarrow R_3 - R_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & 0 & -1 \\ 0 & 1 & -3 & \frac{5}{2} & \frac{1}{2} & -\frac{5}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] \end{aligned}$$



$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & 0 & -1 \\ 0 & 1 & -1 & 5 & 0 & -3 \\ 0 & 0 & 1 & 0 & 1 & 3 \end{array} \right] \leftarrow \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & 0 & -1 \\ 0 & 1 & -1 & 5 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 3 \end{array} \right] \uparrow \underbrace{\phantom{\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & 0 & -1 \\ 0 & 1 & -1 & 5 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 3 \end{array} \right]}}_{E^{-1}}$$

$$E^{-1} = \left[\begin{array}{ccc} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{array} \right]$$

Ili drugi način:

$$[E | I] = \left[\begin{array}{ccc|ccc} 3 & -1 & 1 & 1 & 0 & 0 \\ -15 & 6 & -5 & 0 & 1 & 0 \\ 5 & -2 & 2 & 0 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 3 & -1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 5 & 1 & 0 \\ 0 & -1/3 & 1/3 & -5/3 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 3 & 0 & 1 & 6 & 1 & 0 \\ 0 & 1 & 0 & 5 & 1 & 0 \\ 0 & 0 & 1/3 & 0 & 1/3 & 1 \end{array} \right] \sim$$

$$\sim \left[\begin{array}{ccc|ccc} 3 & 0 & 1 & 6 & 1 & 0 \\ 0 & 1 & 0 & 5 & 1 & 0 \\ 0 & 0 & 1/3 & 0 & 1/3 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 3 & 0 & 0 & 6 & 0 & -3 \\ 0 & 1 & 0 & 5 & 1 & 0 \\ 0 & 0 & 1/3 & 0 & 1/3 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & 0 & -1 \\ 0 & 1 & 0 & 5 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 3 \end{array} \right] \underbrace{\phantom{\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & 0 & -1 \\ 0 & 1 & 0 & 5 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 3 \end{array} \right]}}_{E^{-1}}$$

Objašnjenje:

1. Korak: Prvi redak pomnožili smo s $-\frac{3}{5}$ i dodali trećem i prvi redak pomnožili smo s 5 i dodali drugom.
2. Korak: Drugi redak pridodali smo prvom i trećem retku (poništili smo elemente na poziciji e_{12} i e_{32}).
3. Korak: Treći redak pomnožili smo s -3 i dodali prvom.
4. Korak: Prvi redak podijelili smo s 3 kako bismo dobili na poziciji e_{11} jedinicu, analogno treći redak pomnožili smo s 3 kako bismo dobili na poziciji e_{33} jedinicu.

f) $F = \left[\begin{array}{ccc} 1 & 1 & -3 \\ -1 & 1 & 17 \\ 2 & 3 & 1 \end{array} \right]$

Rješenje:

$$[F | I] = \left[\begin{array}{ccc|ccc} 1 & 1 & -3 & 1 & 0 & 0 \\ -1 & 1 & 17 & 0 & 1 & 0 \\ 2 & 3 & 1 & 0 & 0 & 1 \end{array} \right] \begin{array}{l} \swarrow \cdot (-2) \\ \downarrow \\ \swarrow \end{array} \sim \left[\begin{array}{ccc|ccc} 1 & 1 & -3 & 1 & 0 & 0 \\ 0 & 2 & 14 & 1 & 1 & 0 \\ 0 & 1 & 7 & -2 & 0 & 1 \end{array} \right] \sim$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 1 & -3 & 1 & 0 & 0 \\ 0 & 1 & 7 & -2 & 0 & 1 \\ 0 & 2 & 14 & 1 & 1 & 0 \end{array} \right] \leftarrow \begin{array}{l} \swarrow \cdot (-1) \swarrow \cdot (-2) \\ \downarrow \end{array} \sim \left[\begin{array}{ccc|ccc} 1 & 0 & -10 & 3 & 0 & -1 \\ 0 & 1 & 7 & -2 & 0 & 1 \\ 0 & 0 & 0 & 5 & 1 & -2 \end{array} \right]$$

Vidimo kako matricu s lijeve strane ne možemo svesti na jediničnu iz čega zaključujemo da inverz ne postoji!



AUDITORNE VJEŽBE #9,10 | LINEARNA ALGEBRA | PREDDIPLOMSKI STUDIJ 2020./2021.

Kako bismo bili sigurni, odredimo i determinantu matrice F . Budući da inverz ne postoji, determinanta bi trebala iznositi 0. Uvjerimo se!

$$\det F = \begin{vmatrix} 1 & 1 & -3 \\ -1 & 1 & 17 \\ 2 & 3 & 1 \end{vmatrix} = 1 \cdot \begin{vmatrix} 1 & 17 \\ 3 & 1 \end{vmatrix} - 1 \cdot \begin{vmatrix} -1 & 17 \\ 2 & 1 \end{vmatrix} - 3 \cdot \begin{vmatrix} -1 & 1 \\ 2 & 3 \end{vmatrix} =$$

$$= (1 - 3 \cdot 17) - (-1 \cdot 1 - 2 \cdot 17) - 3 \cdot (-1 \cdot 3 - 2 \cdot 1) = -50 + 35 + 15 = 0$$

Zadatak 4. Odredite inverz matrice X koja je zadana s $X = A + 2B^T$, ako je

$$A = \begin{bmatrix} 3 & 1 & -3 \\ 6 & 1 & 3 \\ 2 & 0 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & -5 & 0 \\ -1 & 2 & -3 \\ 3 & -6 & -2 \end{bmatrix}.$$

Rješenje:

$$B^T = \begin{bmatrix} -1 & -1 & 3 \\ -5 & 2 & -6 \\ 0 & -3 & -2 \end{bmatrix}$$

$$X = A + 2B^T = \begin{bmatrix} 3 & 1 & -3 \\ 6 & 1 & 3 \\ 2 & 0 & -1 \end{bmatrix} + 2 \cdot \begin{bmatrix} -1 & -1 & 3 \\ -5 & 2 & -6 \\ 0 & -3 & -2 \end{bmatrix} = \begin{bmatrix} 3 & 1 & -3 \\ 6 & 1 & 3 \\ 2 & 0 & -1 \end{bmatrix} + \begin{bmatrix} -2 & -2 & 6 \\ -10 & 4 & -12 \\ 0 & -6 & -4 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 3 \\ -4 & 5 & -9 \\ 2 & -6 & -5 \end{bmatrix}$$

$$[X | I] = \left[\begin{array}{ccc|ccc} 1 & -1 & 3 & 1 & 0 & 0 \\ -4 & 5 & -9 & 0 & 1 & 0 \\ 2 & -6 & -5 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\substack{/\cdot 4 \quad /\cdot (-2) \\ \downarrow \quad \downarrow}} \sim \left[\begin{array}{ccc|ccc} 1 & -1 & 3 & 1 & 0 & 0 \\ 0 & 1 & 3 & 4 & 1 & 0 \\ 0 & -4 & -11 & -2 & 0 & 1 \end{array} \right] \xrightarrow{/\cdot 4 \sim} \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 6 & 5 & 1 & 0 \\ 0 & 1 & 3 & 4 & 1 & 0 \\ 0 & 0 & 1 & 14 & 4 & 1 \end{array} \right] \xrightarrow{\substack{\leftarrow \quad \leftarrow \\ \leftarrow \quad \leftarrow}} \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -79 & -23 & -6 \\ 0 & 1 & 0 & -38 & -11 & -3 \\ 0 & 0 & 1 & 14 & 4 & 1 \end{array} \right] \xrightarrow{/\cdot (-3) \quad /\cdot (-6)}$$

$$X^{-1} = \begin{bmatrix} -79 & -23 & -6 \\ -38 & -11 & -3 \\ 14 & 4 & 1 \end{bmatrix}$$

Zadatak 5. Odredite inverz i trag matrice X koja je zadana s $X = (A + 3I) \cdot C - B$, ako je

$$A = \begin{bmatrix} -2 & 3 \\ 1 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 5 \\ 1 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 4 & 1 \\ 0 & 2 \end{bmatrix} \text{ i } I \text{ je jedinična matrica dimenzije } 2 \times 2.$$

Rješenje:

$$A + 3I = \begin{bmatrix} -2 & 3 \\ 1 & -1 \end{bmatrix} + 3 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 3 \\ 1 & -1 \end{bmatrix} + \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 1 & 2 \end{bmatrix}$$

$$(A + 3I) \cdot C = \begin{bmatrix} 1 & 3 \\ 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 4 & 1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 7 \\ 4 & 5 \end{bmatrix}$$



$$X = (A + 3I) \cdot C - B = \begin{bmatrix} 4 & 7 \\ 4 & 5 \end{bmatrix} - \begin{bmatrix} 3 & 5 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$$

$$\text{tr } X = 1 + 5 = 6$$

$$[X | I] = \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 5 & 0 & 1 \end{array} \right] \xrightarrow{\cdot(-3)} \sim \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -1 & -3 & 1 \end{array} \right] \xleftarrow{\cdot 2} \sim \left[\begin{array}{cc|cc} 1 & 0 & -5 & 2 \\ 0 & -1 & -3 & 1 \end{array} \right] \xrightarrow{\cdot(-1)} \sim \left[\begin{array}{cc|cc} 1 & 0 & -5 & 2 \\ 0 & 1 & 3 & -1 \end{array} \right]$$

$$X^{-1} = \begin{bmatrix} -5 & 2 \\ 3 & -1 \end{bmatrix}$$

Možemo odrediti inverz matrice pomoću matrice kofaktora:

$$\det X = \begin{vmatrix} 1 & 2 \\ 3 & 5 \end{vmatrix} = 1 \cdot 5 - 2 \cdot 3 = -1.$$

$$A_{11} = (-1)^{1+1} \cdot 5 = 5 \quad (\text{uklonili prvi redak i prvi stupac})$$

$$A_{12} = (-1)^{1+2} \cdot 3 = -3 \quad (\text{uklonili prvi redak i drugi stupac})$$

$$A_{21} = (-1)^{2+1} \cdot 2 = -2 \quad (\text{uklonili drugi redak i prvi stupac})$$

$$A_{22} = (-1)^{2+2} \cdot 1 = 1 \quad (\text{uklonili drugi redak i drugi stupac})$$

$$A^* = \begin{bmatrix} 5 & -3 \\ -2 & 1 \end{bmatrix}$$

Inverz matrice A je:

$$\Rightarrow A^{-1} = \frac{1}{\det A} (A^*)^T = \frac{1}{-1} \begin{bmatrix} 5 & -2 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} -5 & 2 \\ 3 & -1 \end{bmatrix}.$$

Zadatak 6: Ispitajte regularnost matrice $C = (A \cdot B)^T + (B - 2I)^{-1}$, ako je $A = \begin{bmatrix} 2 & -1 & 2 \\ 0 & 3 & 1 \\ -2 & 5 & 2 \end{bmatrix}$,

$$B = \begin{bmatrix} 3 & -2 & 0 \\ 3 & -3 & 2 \\ -2 & 4 & 3 \end{bmatrix} \text{ i } I \text{ je jedinična matrica dimenzije } 3 \times 3.$$

Rješenje:

$$\begin{aligned} A \cdot B &= \begin{bmatrix} 2 & -1 & 2 \\ 0 & 3 & 1 \\ -2 & 5 & 2 \end{bmatrix} \cdot \begin{bmatrix} 3 & -2 & 0 \\ 3 & -3 & 2 \\ -2 & 4 & 3 \end{bmatrix} = \begin{bmatrix} 2 \cdot 3 + (-1) \cdot 3 + 2 \cdot (-2) & 2 \cdot (-2) + (-1) \cdot (-3) + 2 \cdot 4 & 2 \cdot 0 + (-1) \cdot 2 + 2 \cdot 3 \\ 0 \cdot 3 + 3 \cdot 3 + 1 \cdot (-2) & 0 \cdot (-2) + 3 \cdot (-3) + 1 \cdot 4 & 0 \cdot 0 + 3 \cdot 2 + 1 \cdot 3 \\ -2 \cdot 3 + 5 \cdot 3 + 2 \cdot (-2) & (-2) \cdot (-2) + 5 \cdot (-3) + 2 \cdot 4 & (-2) \cdot 0 + 5 \cdot 2 + 2 \cdot 3 \end{bmatrix} = \\ &= \begin{bmatrix} -1 & 7 & 4 \\ 7 & -5 & 9 \\ 5 & -3 & 16 \end{bmatrix} \end{aligned}$$



$$(A \cdot B)^T = \begin{bmatrix} -1 & 7 & 5 \\ 7 & -5 & -3 \\ 4 & 9 & 16 \end{bmatrix}$$

$$B - 2I = \begin{bmatrix} 3 & -2 & 0 \\ 3 & -3 & 2 \\ -2 & 4 & 3 \end{bmatrix} - \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & -2 & 0 \\ 3 & -5 & 2 \\ -2 & 4 & 1 \end{bmatrix}$$

Odredimo inverz matrice $B - 2I$:

$$\begin{bmatrix} 1 & -2 & 0 & 1 & 0 & 0 \\ 3 & -5 & 2 & 0 & 1 & 0 \\ -2 & 4 & 1 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\substack{/\cdot(-3) \quad /\cdot 2 \\ \swarrow \quad \downarrow}} \sim \begin{bmatrix} 1 & -2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & -3 & 1 & 0 \\ 0 & 0 & 1 & 2 & 0 & 1 \end{bmatrix} \xleftarrow{\substack{/\cdot 2 \\ \swarrow \quad \downarrow}} \begin{bmatrix} 1 & 0 & 4 & -5 & 2 & 0 \\ 0 & 1 & 2 & -3 & 1 & 0 \\ 0 & 0 & 1 & 2 & 0 & 1 \end{bmatrix} \xleftarrow{\substack{/\cdot(-2) \quad /\cdot(-4)}} \sim \begin{bmatrix} 1 & 0 & 0 & -13 & 2 & -4 \\ 0 & 1 & 0 & -7 & 1 & -2 \\ 0 & 0 & 1 & 2 & 0 & 1 \end{bmatrix}$$

$$\rightarrow (B - 2I)^{-1} = \begin{bmatrix} -13 & 2 & -4 \\ -7 & 1 & -2 \\ 2 & 0 & 1 \end{bmatrix}$$

$$C = (A \cdot B)^T + (B - 2I)^{-1} = \begin{bmatrix} -1 & 7 & 5 \\ 7 & -5 & -3 \\ 4 & 9 & 16 \end{bmatrix} + \begin{bmatrix} -13 & 2 & -4 \\ -7 & 1 & -2 \\ 2 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -14 & 9 & 1 \\ 0 & -4 & -5 \\ 6 & 9 & 17 \end{bmatrix}$$

Matrica je regularna ukoliko je determinanta te matrice različita od 0. Odredimo stoga determinantu matrice C :

$$\det C = \begin{vmatrix} -14 & 9 & 1 \\ 0 & -4 & -5 \\ 6 & 9 & 17 \end{vmatrix} = -14 \cdot \begin{vmatrix} -4 & -5 \\ 9 & 17 \end{vmatrix} + 6 \cdot \begin{vmatrix} 9 & 1 \\ -4 & -5 \end{vmatrix} = -14 \cdot (-4 \cdot 17 - 9 \cdot (-5)) + 6 \cdot (9 \cdot (-5) - (-4) \cdot 1) =$$

$$= -14 \cdot (-23) + 6 \cdot (-41) = 76 \neq 0$$

Zaključujemo da je matrica C regularna!



III. 5. Rang matrice

Definicija 2. Broj linearno nezavisnih redaka matrice $A \in M_{mn}$ jednak je broju linearno nezavisnih stupaca i naziva se **rang matrice** A .

Za određivanje ranga matrice dovedemo matricu elementarnim transformacijama do toga da ispod „glavne dijagonale“ (tj. ispod elemenata $a_{11}, a_{22}, a_{33}, \dots$) imamo sve nule i onda je rang matrice jednak broju redaka različitih od nule!

Prethodna činjenica posljedica je toga što **ekvivalentne matrice imaju jednaki rang**.

Iz svega navedenog možemo zaključiti sljedeće:

Napomena 1. Matrica $A \in M_n$ je regularna ako i samo ako je $r(A) = n$.

$$r(A) \leq \min\{m, n\} \quad (A \in M_{mn})$$

(Objašnjenje: Prethodna napomena je skroz očita, jer matrica je regularna ako postoji inverz. A ukoliko krenemo vršiti elementarne transformacije nad redcima i uspijemo dobiti jediničnu matricu s lijeve strane tada je rang matrice jednak broju linearno nezavisnih redaka, a to je upravo broj redaka matrice A . Ukoliko dobijemo da je jedan redak nul-redak – pogledajte zadatak 3. f) tada ne možemo odrediti inverz, a rang matrice je manji od broja redaka matrice A . U tom zadatku 3. f), $r(F) = 3 < 4 = n$.)

Zadatak 7. Odredite rang sljedećih matrica:

$$\text{a) } A = \begin{bmatrix} 1 & 2 & 1 & 1 \\ 2 & 0 & 1 & -1 \\ 0 & 0 & 2 & 0 \end{bmatrix}$$

Rješenje:

$$\begin{bmatrix} 1 & 2 & 1 & 1 \\ 2 & 0 & 1 & -1 \\ 0 & 0 & 2 & 0 \end{bmatrix} \xrightarrow{\substack{R_2 \leftarrow R_2 - 2R_1 \\ R_3 \leftarrow R_3 \cdot (-2)}} \sim \begin{bmatrix} 1 & 2 & 1 & 1 \\ 0 & -4 & -1 & -3 \\ 0 & 0 & 2 & 0 \end{bmatrix} \Rightarrow r(A) = 3$$

(Napomena: Mogli ste matricu svoditi da dobijete maksimalnu jediničnu matricu:

$$\begin{bmatrix} 1 & 2 & 1 & 1 \\ 2 & 0 & 1 & -1 \\ 0 & 0 & 2 & 0 \end{bmatrix} \xrightarrow{\substack{R_2 \leftarrow R_2 - 2R_1 \\ R_3 \leftarrow R_3 \cdot (-2)}} \sim \begin{bmatrix} 1 & 2 & 1 & 1 \\ 0 & -4 & -1 & -3 \\ 0 & 0 & 2 & 0 \end{bmatrix} \xrightarrow{\substack{R_2 \leftarrow R_2 \cdot (-1/4) \\ R_3 \leftarrow R_3 \cdot (1/2)}} \sim \begin{bmatrix} 1 & 2 & 1 & 1 \\ 0 & -1 & 1/4 & 3/4 \\ 0 & 0 & 1 & 0 \end{bmatrix} \xrightarrow{\substack{R_1 \leftarrow R_1 - 2R_2 \\ R_2 \leftarrow R_2 \cdot (-4)}} \sim \begin{bmatrix} 1 & 0 & 1/2 & -1/2 \\ 0 & -1 & 1/4 & 3/4 \\ 0 & 0 & 1 & 0 \end{bmatrix} \xrightarrow{\substack{R_1 \leftarrow R_1 - (1/2)R_3 \\ R_2 \leftarrow R_2 \cdot (-4)}} \sim \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \Rightarrow r(A) = 3$$

(Napomena: Ukoliko prvi stupac pomnožimo s $\frac{1}{2}$ i dodamo 3. stupcu; i ukoliko 2. stupac pomnožimo s $-\frac{3}{4}$ i dodamo 3. stupcu dobivamo ekvivalentnu matricu:



$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \text{ iz koje uistinu uočavamo da je najveća jedinična matrica dimenzije } 3 \times 3.)$$

$$\text{b) } B = \begin{bmatrix} 2 & 3 & -1 & 4 \\ 1 & 0 & 1 & 2 \\ 3 & 4 & 0 & 7 \\ -2 & -1 & 4 & 1 \\ 4 & -2 & 3 & 5 \end{bmatrix}$$

Rješenje:

$$\begin{bmatrix} 2 & 3 & -1 & 4 \\ 1 & 0 & 1 & 2 \\ 3 & 4 & 0 & 7 \\ -2 & -1 & 4 & 1 \\ 4 & -2 & 3 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 2 \\ 2 & 3 & -1 & 4 \\ 3 & 4 & 0 & 7 \\ -2 & -1 & 4 & 1 \\ 4 & -2 & 3 & 5 \end{bmatrix} \begin{array}{l} \swarrow \cdot (-2) \quad \swarrow \cdot (-3) \quad \swarrow \cdot 2 \quad \swarrow \cdot (-4) \\ \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\ \swarrow \quad \downarrow \quad \downarrow \\ \downarrow \quad \downarrow \\ \swarrow \end{array} \sim$$

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 3 & -3 & 0 \\ 0 & 4 & -3 & 1 \\ 0 & -1 & 6 & 5 \\ 0 & -2 & -1 & -3 \end{bmatrix} \begin{array}{l} \swarrow :3 \\ \downarrow \end{array} \sim \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & -1 & 0 \\ 0 & 4 & -3 & 1 \\ 0 & -1 & 6 & 5 \\ 0 & -2 & -1 & -3 \end{bmatrix} \begin{array}{l} \swarrow \cdot (-4) \quad \downarrow \quad \swarrow :2 \\ \swarrow \quad \downarrow \quad \downarrow \quad \sim \\ \swarrow \quad \downarrow \\ \swarrow \end{array}$$

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 5 & 5 \\ 0 & 0 & -3 & -3 \end{bmatrix} \begin{array}{l} \swarrow \cdot (-5) \quad \swarrow :3 \\ \downarrow \quad \downarrow \end{array} \sim \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{array}{l} \swarrow \cdot (-1) \quad \uparrow \\ \uparrow \quad \uparrow \end{array} \sim \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow r(B) = 3$$

(Napomena: Ukoliko 1. stupac pomnožimo s -1 i dodamo 4. stupcu; i ukoliko 2. stupac pomnožimo s -1 i dodamo 4. stupcu; i ukoliko 3. stupac pomnožimo s -1 i dodamo 4. stupcu dobivamo ekvivalentnu matricu:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ iz koje uistinu uočavamo da je najveća jedinična matrica dimenzije } 3 \times 3)$$

Zadatak 8. U ovisnosti o realnom parametru $\lambda \in \mathbb{R}$ odredite rang matrice:

$$A = \begin{bmatrix} 2 & 3 & 1 & \lambda \\ 2 & \lambda & 3 & 5 \\ 3 & 4 & 5 & 4 \\ 1 & 1 & 2 & 2 \end{bmatrix}$$



Rješenje:

$$\begin{bmatrix} 2 & 3 & 1 & \lambda \\ 2 & \lambda & 3 & 5 \\ 3 & 4 & 5 & 4 \\ 1 & 1 & 2 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 & 2 \\ 2 & \lambda & 3 & 5 \\ 3 & 4 & 5 & 4 \\ 2 & 3 & 1 & \lambda \end{bmatrix} \begin{array}{l} \leftarrow \\ \downarrow \\ \downarrow \\ \downarrow \end{array} \begin{array}{l} / \cdot (-2) \quad / \cdot (-3) \quad / \cdot (-2) \\ \\ \\ \end{array} \sim$$

$$\sim \begin{bmatrix} 1 & 1 & 2 & 2 \\ 0 & \lambda-2 & -1 & 1 \\ 0 & 1 & -1 & -2 \\ 0 & 1 & -3 & \lambda-4 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 & 2 \\ 0 & 1 & -1 & -2 \\ 0 & \lambda-2 & -1 & 1 \\ 0 & 1 & -3 & \lambda-4 \end{bmatrix} \begin{array}{l} \leftarrow \\ \\ \downarrow \\ \leftarrow \end{array} \begin{array}{l} / \cdot (-1) \quad / \cdot (-(\lambda-2)) \quad / \cdot (-1) \\ \\ \\ \end{array} \sim$$

$$\sim \begin{bmatrix} 1 & 0 & 3 & 4 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & \lambda-3 & 2\lambda-3 \\ 0 & 0 & -2 & \lambda-2 \end{bmatrix} / : (-2) \sim \begin{bmatrix} 1 & 0 & 3 & 4 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 1 & \frac{2-\lambda}{2} \\ 0 & 0 & \lambda-3 & 2\lambda-3 \end{bmatrix} \begin{array}{l} \leftarrow \\ \uparrow \\ \leftarrow \\ \downarrow \end{array} \begin{array}{l} \\ / \cdot (-3) \quad \uparrow \quad / \cdot (-(\lambda-3)) \\ \\ \end{array} \sim$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & \frac{3\lambda+6}{2} \\ 0 & 1 & 0 & -\frac{\lambda+2}{2} \\ 0 & 0 & 1 & \frac{2-\lambda}{2} \\ 0 & 0 & 0 & \frac{\lambda^2-\lambda}{2} \end{bmatrix}$$

Zaključujemo sljedeće:

- Ukoliko je $\frac{\lambda^2-\lambda}{2} = 0$ tada je posljednji redak nul-redak, tj. $r(A) = 3$.
- Ukoliko je $\frac{\lambda^2-\lambda}{2} \neq 0$ tada dijeljenjem posljednjeg retka s $\frac{\lambda^2-\lambda}{2}$ dobivamo redak koji ima sve nule, a na posljednjem mjestu 1, tj. zaključujemo da je $r(A) = 4$.

$$\frac{\lambda^2-\lambda}{2} = 0 \Rightarrow \lambda_1 = 0, \lambda_2 = 1$$

Konačno:

- Za $\lambda \in \{0, 1\}$ je $r(A) = 3$;
- Za $\lambda \in \mathbb{R} \setminus \{0, 1\}$ je $r(A) = 4$.



III. 6. Matrične jednadžbe

Promotrimo 2 tipa matričnih jednadžbi: $AX = B$ i $YA = C$ pri čemu je množenje matrica dobro definirano.

Odnosno, vrijedi:

- $A \in M_{mn}$, $X \in M_{nr}$, $B \in M_{mr}$ i
- $Y \in M_{rm}$, $C \in M_{rn}$.

Ukoliko želimo odrediti X i Y sjetimo se da je $A \cdot A^{-1} = A^{-1} \cdot A = I$. Stoga:

Množenje s lijeva:

$$A^{-1} \cdot AX = B$$

$$A^{-1}AX = A^{-1}B$$

$$I \cdot X = A^{-1}B$$

$$X = A^{-1}B$$

Množenje s desna:

$$YA = C \quad /A^{-1}$$

$$YAA^{-1} = CA^{-1}$$

$$Y \cdot I = CA^{-1}$$

$$Y = CA^{-1}$$

Budući da za množenje matrica ne vrijedi komutativnost primijetimo kako nije svejedno množimo li jednadžbu s lijeva ili s desna.

(Primjedba: Što bi se dogodilo da smo slučaj $AX = B$ pokušali riješiti množeći jednadžbu s A^{-1} s desne strane? Tada bi jednadžba bila oblika: $AXA^{-1} = BA^{-1}$, a kako ne smijemo zamijeniti redoslijed matrica X i A^{-1} (jer množenje matrica nije komutativno) ne možemo poništiti matricu A kako bi ostala sama matrica X s lijeve strane. Stoga je jako važno s koje strane množimo jednadžbu matricom.)

Zadatak 9. Riješite matričnu jednadžbu $XA = B$ ako je

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 1 \\ -1 & 1 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & -2 & 3 \\ 1 & 0 & 4 \end{bmatrix}.$$

Rješenje:

$$XA = B \quad /A^{-1}$$

$$XAA^{-1} = BA^{-1}$$

$$X \cdot I = BA^{-1}$$

$$X = BA^{-1}$$

A^{-1} možemo odrediti Gauss – Jordanovom metodom za određivanje inverza:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ -1 & 1 & 0 & 0 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 2 & 1 & 0 & 1 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & -1 & -2 & 1 & -2 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 1 & -2 \\ 0 & 1 & 0 & -1 & 1 & -1 \\ 0 & 0 & -1 & -2 & 1 & -2 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 1 & -2 \\ 0 & 1 & 0 & -1 & 1 & -1 \\ 0 & 0 & 1 & 2 & -1 & 2 \end{array} \right] \Rightarrow A^{-1} = \begin{bmatrix} -1 & 1 & -2 \\ -1 & 1 & -1 \\ 2 & -1 & 2 \end{bmatrix}$$



I onda je

$$X = \begin{bmatrix} 0 & -2 & 3 \\ 1 & 0 & 4 \end{bmatrix} \cdot \begin{bmatrix} -1 & 1 & -2 \\ -1 & 1 & -1 \\ 2 & -1 & 2 \end{bmatrix}$$

$$X = \begin{bmatrix} 0 \cdot (-1) + (-2) \cdot (-1) + 3 \cdot 2 & 0 \cdot 1 + (-2) \cdot 1 + 3 \cdot (-1) & 0 \cdot (-2) + (-2) \cdot (-1) + 3 \cdot 2 \\ 1 \cdot (-1) + 0 \cdot (-1) + 4 \cdot 2 & 1 \cdot 1 + 0 \cdot 1 + 4 \cdot (-1) & 1 \cdot (-2) + 0 \cdot (-1) + 4 \cdot 2 \end{bmatrix}$$

$$X = \begin{bmatrix} 8 & -5 & 8 \\ 7 & -3 & 6 \end{bmatrix}$$

Zadatak 10. Riješite jednadžbu $AX = B$ ako je $A = \begin{bmatrix} 1 & -2 \\ 5 & -9 \end{bmatrix}$ i $B = \begin{bmatrix} -2 & 3 \\ 0 & -5 \end{bmatrix}$. Koliko iznosi trag matrice X ?

Rješenje:

Prisjetimo se da množenje matrica NIJE komutativno, stoga nije isto množimo li cijelu jednadžbu s lijeva ili s desna. Ukoliko želimo doći do matrice X tada moramo "poništiti" matricu A , tj. vrijedi $A^{-1} \cdot A = A \cdot A^{-1} = I$. Stoga moramo množiti cijelu jednadžbu s lijeva matricom A^{-1} :

$$A^{-1} \cdot AX = B$$

$$A^{-1}AX = A^{-1}B$$

$$X = A^{-1}B$$

Odredimo inverz matrice A , to možemo učiniti pomoću matrice kofaktora ili Gauss – Jordanovom metodom eliminacije.

i) Inverz matrice pomoću adjunkte:

$$A^{-1} = \frac{1}{\det A} (A^*)^T$$

→ $A_{ij} = (-1)^{i+j} \det M_{ij}$ (Matricu M_{ij} dobijemo tako da uklonimo i-ti redak i j-ti stupac matrice A)

$$\det A = \begin{vmatrix} 1 & -2 \\ 5 & -9 \end{vmatrix} = -9 - (-10) = 1$$

$$A_{11} = (-1)^{1+1} \cdot (-9) = -9 \text{ (uklonimo prvi redak i prvi stupac matrice A)}$$

$$A_{12} = (-1)^{1+2} \cdot 5 = -5$$

$$A_{21} = (-1)^{2+1} \cdot (-2) = 2$$

$$A_{22} = (-1)^{2+2} \cdot 1 = 1$$

$$A^* = \begin{bmatrix} -9 & -5 \\ 2 & 1 \end{bmatrix}, \text{ pa je inverz matrice}$$

$$A^{-1} = \frac{1}{\det A} (A^*)^T = \frac{1}{1} \cdot \left(\begin{bmatrix} -9 & -5 \\ 2 & 1 \end{bmatrix} \right)^T = \begin{bmatrix} -9 & 2 \\ -5 & 1 \end{bmatrix}$$

Općenito, za matricu 2×2 koja je oblika $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ inverz je jednak



$$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

ii) Gauss - Jordanova metoda eliminacije

$$\left[\begin{array}{cc|cc} 1 & -2 & 1 & 0 \\ 5 & -9 & 0 & 1 \end{array} \right] \xrightarrow{\cdot(-5)} \sim \left[\begin{array}{cc|cc} 1 & -2 & 1 & 0 \\ 0 & 1 & -5 & 1 \end{array} \right] \xleftarrow{\cdot 2} \sim \left[\begin{array}{cc|cc} 1 & 0 & -9 & 2 \\ 0 & 1 & -5 & 1 \end{array} \right]$$

$$\rightarrow A^{-1} = \begin{bmatrix} -9 & 2 \\ -5 & 1 \end{bmatrix}$$

$$X = A^{-1}B = \begin{bmatrix} -9 & 2 \\ -5 & 1 \end{bmatrix} \cdot \begin{bmatrix} -2 & 3 \\ 0 & -5 \end{bmatrix} = \begin{bmatrix} (-9) \cdot (-2) + 2 \cdot 0 & -9 \cdot 3 + 2 \cdot (-5) \\ (-5) \cdot (-2) + 1 \cdot 0 & -5 \cdot 3 + 1 \cdot (-5) \end{bmatrix}$$

$$X = \begin{bmatrix} 18 & -37 \\ 10 & -20 \end{bmatrix}$$

Trag matrice dobijemo tako da zbrojimo elemente na glavnoj dijagonali:

$$\text{tr } X = 18 + (-20) = -2.$$

Zadatak 11. Riješite jednačinu $BX + A = 3I - X$ ako je $A = \begin{bmatrix} 4 & 1 & -1 \\ -3 & -2 & 2 \\ 1 & 0 & 3 \end{bmatrix}$ i $B = \begin{bmatrix} 0 & 2 & 3 \\ -2 & -6 & -8 \\ 3 & 2 & 1 \end{bmatrix}$.

Rješenje:

$$BX + A = 3I - X$$

$$BX + X = 3I - A$$

$$(B+I)^{-1} \cdot (B+I)X = 3I - A$$

$$X = (B+I)^{-1} \cdot (3I - A)$$

$$3I - A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} - \begin{bmatrix} 4 & 1 & -1 \\ -3 & -2 & 2 \\ 1 & 0 & 3 \end{bmatrix} = \begin{bmatrix} -1 & -1 & 1 \\ 3 & 5 & -2 \\ -1 & 0 & 0 \end{bmatrix}$$

$$B+I = \begin{bmatrix} 0 & 2 & 3 \\ -2 & -6 & -8 \\ 3 & 2 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ -2 & -5 & -8 \\ 3 & 2 & 2 \end{bmatrix}$$

$$[B+I | I] = \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ -2 & -5 & -8 & 0 & 1 & 0 \\ 3 & 2 & 2 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\cdot 2, \cdot (-3)} \sim \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & -1 & -2 & 2 & 1 & 0 \\ 0 & -4 & -7 & -3 & 0 & 1 \end{array} \right] \xrightarrow{\cdot (-1)} \sim$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 2 & -2 & -1 & 0 \\ 0 & -4 & -7 & -3 & 0 & 1 \end{array} \right] \xleftarrow{\cdot (-2)} \sim \left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 5 & 2 & 0 \\ 0 & 1 & 2 & -2 & -1 & 0 \\ 0 & 0 & 1 & -11 & -4 & 1 \end{array} \right] \xleftarrow{\cdot (-2)} \sim$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -6 & -2 & 1 \\ 0 & 1 & 0 & 20 & 7 & -2 \\ 0 & 0 & 1 & -11 & -4 & 1 \end{array} \right] \xleftarrow{\cdot (-1)} \sim$$



$$(B+I)^{-1} = \begin{bmatrix} -6 & -2 & 1 \\ 20 & 7 & -2 \\ -11 & -4 & 1 \end{bmatrix}$$

$$X = (B+I)^{-1} \cdot (3I - A) = \begin{bmatrix} -6 & -2 & 1 \\ 20 & 7 & -2 \\ -11 & -4 & 1 \end{bmatrix} \cdot \begin{bmatrix} -1 & -1 & 1 \\ 3 & 5 & -2 \\ -1 & 0 & 0 \end{bmatrix} =$$

$$= \begin{bmatrix} -6 \cdot (-1) + (-2) \cdot 3 + 1 \cdot (-1) & -6 \cdot (-1) + (-2) \cdot 5 + 1 \cdot 0 & -6 \cdot 1 + (-2) \cdot (-2) + 1 \cdot 0 \\ 20 \cdot (-1) + 7 \cdot 3 + (-2) \cdot (-1) & 20 \cdot (-1) + 7 \cdot 5 + (-2) \cdot 0 & 20 \cdot 1 + 7 \cdot (-2) + (-2) \cdot 0 \\ -11 \cdot (-1) + (-4) \cdot 3 + 1 \cdot (-1) & -11 \cdot (-1) + (-4) \cdot 5 + 1 \cdot 0 & -11 \cdot 1 + (-4) \cdot (-2) + 1 \cdot 0 \end{bmatrix}$$

$$X = \begin{bmatrix} -1 & -4 & -2 \\ 3 & 15 & 6 \\ -2 & -9 & -3 \end{bmatrix}$$

Zadatak 12. Riješite jednadžbu $2X - B^2 = AX + I$ ako je $A = \begin{bmatrix} 1 & 4 & -2 \\ 2 & -5 & 3 \\ -4 & 10 & -1 \end{bmatrix}$ i $B = \begin{bmatrix} -1 & 0 & 1 \\ 0 & 2 & 1 \\ -2 & 3 & 1 \end{bmatrix}$.

Rješenje:

$$2X - B^2 = AX + I$$

$$2X - AX = B^2 + I$$

$$(2I - A)^{-1} \cdot (2I - A)X = B^2 + I$$

$$X = (2I - A)^{-1} \cdot (B^2 + I)$$

$$B^2 = B \cdot B = \begin{bmatrix} -1 & 0 & 1 \\ 0 & 2 & 1 \\ -2 & 3 & 1 \end{bmatrix} \cdot \begin{bmatrix} -1 & 0 & 1 \\ 0 & 2 & 1 \\ -2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} -1 \cdot (-1) + 0 \cdot 0 + 1 \cdot (-2) & -1 \cdot 0 + 0 \cdot 2 + 1 \cdot 3 & -1 \cdot 1 + 0 \cdot 1 + 1 \cdot 1 \\ 0 \cdot (-1) + 2 \cdot 0 + 1 \cdot (-2) & 0 \cdot 0 + 2 \cdot 2 + 1 \cdot 3 & 0 \cdot 1 + 2 \cdot 1 + 1 \cdot 1 \\ -2 \cdot (-1) + 3 \cdot 0 + 1 \cdot (-2) & -2 \cdot 0 + 3 \cdot 2 + 1 \cdot 3 & -2 \cdot 1 + 3 \cdot 1 + 1 \cdot 1 \end{bmatrix}$$

$$B^2 = \begin{bmatrix} -1 & 3 & 0 \\ -2 & 7 & 3 \\ 0 & 9 & 2 \end{bmatrix}$$

$$B^2 + I = \begin{bmatrix} -1 & 3 & 0 \\ -2 & 7 & 3 \\ 0 & 9 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 3 & 0 \\ -2 & 8 & 3 \\ 0 & 9 & 3 \end{bmatrix}$$

$$2I - A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 4 & -2 \\ 2 & -5 & 3 \\ -4 & 10 & -1 \end{bmatrix} = \begin{bmatrix} 1 & -4 & 2 \\ -2 & 7 & -3 \\ 4 & -10 & 3 \end{bmatrix}$$



AUDITORNE VJEŽBE #9,10 | LINEARNA ALGEBRA | PREDDIPLOMSKI STUDIJ 2020./2021.

$$\begin{aligned}
 [2I - A | I] &= \left[\begin{array}{ccc|ccc} 1 & -4 & 2 & 1 & 0 & 0 \\ -2 & 7 & -3 & 0 & 1 & 0 \\ 4 & -10 & 3 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\substack{/\cdot 2 \quad /\cdot (-4) \\ \downarrow \\ \downarrow}} \sim \left[\begin{array}{ccc|ccc} 1 & -4 & 2 & 1 & 0 & 0 \\ 0 & -1 & 1 & 2 & 1 & 0 \\ 0 & 6 & -5 & -4 & 0 & 1 \end{array} \right] \xrightarrow{/\cdot (-1)} \sim \\
 &\sim \left[\begin{array}{ccc|ccc} 1 & -4 & 2 & 1 & 0 & 0 \\ 0 & 1 & -1 & -2 & -1 & 0 \\ 0 & 6 & -5 & -4 & 0 & 1 \end{array} \right] \xleftarrow{/\cdot 4 \quad /\cdot (-6)} \sim \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & -7 & -4 & 0 \\ 0 & 1 & -1 & -2 & -1 & 0 \\ 0 & 0 & 1 & 8 & 6 & 1 \end{array} \right] \xleftarrow{\uparrow \cdot 2} \sim \\
 &\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 9 & 8 & 2 \\ 0 & 1 & 0 & 6 & 5 & 1 \\ 0 & 0 & 1 & 8 & 6 & 1 \end{array} \right] \xrightarrow{\substack{\underbrace{\hspace{1cm}}_{(2I-A)^{-1}}}}
 \end{aligned}$$

$$(2I - A)^{-1} = \begin{bmatrix} 9 & 8 & 2 \\ 6 & 5 & 1 \\ 8 & 6 & 1 \end{bmatrix}$$

$$X = (2I - A)^{-1} \cdot (B^2 + I) = \begin{bmatrix} 9 & 8 & 2 \\ 6 & 5 & 1 \\ 8 & 6 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 & 3 & 0 \\ -2 & 8 & 3 \\ 0 & 9 & 3 \end{bmatrix} = \begin{bmatrix} 9 \cdot 0 + 8 \cdot (-2) + 2 \cdot 0 & 9 \cdot 3 + 8 \cdot 8 + 2 \cdot 9 & 9 \cdot 0 + 8 \cdot 3 + 2 \cdot 3 \\ 6 \cdot 0 + 5 \cdot (-2) + 1 \cdot 0 & 6 \cdot 3 + 5 \cdot 8 + 1 \cdot 9 & 6 \cdot 0 + 5 \cdot 3 + 1 \cdot 3 \\ 8 \cdot 0 + 6 \cdot (-2) + 1 \cdot 0 & 8 \cdot 3 + 6 \cdot 8 + 1 \cdot 9 & 8 \cdot 0 + 6 \cdot 3 + 1 \cdot 3 \end{bmatrix}$$

$$X = \begin{bmatrix} -16 & 109 & 30 \\ -10 & 67 & 18 \\ -12 & 81 & 21 \end{bmatrix}$$

**Zadaci za samostalan rad:**

1. Odredite $x \in \mathbb{R}$ za koje je matrica A regularna.

$$A = \begin{bmatrix} x^2 & 4 & 18 & 0 \\ x & 0 & 3 & -1 \\ 1 & 1 & 1 & 2 \\ 2 & 1 & -1 & 2 \end{bmatrix}$$

Rješenje: $x \in \mathbb{R} \setminus \{-1, 9\}$

2. Za koje vrijednosti realnog parametra λ će rang matrice A biti:

- a) $r(A) = 1$,
- b) $r(A) = 2$,
- c) $r(A) = 3$?

$$A = \begin{bmatrix} -1 & 2 & 1 \\ 2 & \lambda & -2 \\ 3 & -6 & -3 \end{bmatrix}.$$

Rješenje: $r(A) = 1$ je za $\lambda = -4$, $r(A) = 2$ za $\lambda \neq -4$, $r(A) \neq 3$ uvijek!

3. Riješite matričnu jednadžbu $AX = B(X + A)$ ako je:

$$A = \begin{bmatrix} 2 & 0 & -2 \\ 1 & 3 & 1 \\ -1 & 2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -2 & 1 \\ -2 & 1 & 5 \\ -3 & 3 & 1 \end{bmatrix}.$$

Rješenje: $X = \begin{bmatrix} -12 & 33 & 22 \\ -20 & 55 & 34 \\ -17 & 49 & 31 \end{bmatrix}$

4. Ukoliko postoji, odredite inverz matrice

$$A = \begin{bmatrix} 2 & 1 & 5 \\ 6 & 2 & -1 \\ -2 & 3 & 1 \end{bmatrix}$$

Rješenje: $A^{-1} = \frac{1}{119} \begin{bmatrix} 5 & 14 & -11 \\ -4 & 12 & 32 \\ 22 & -8 & -2 \end{bmatrix}.$

5. Ispitajte zadovoljava li rang matrice

$$A = \begin{bmatrix} -1 & 2 & 0 & 1 & 4 \\ 1 & 2 & 3 & -1 & 5 \\ 0 & 4 & 3 & 0 & 9 \end{bmatrix}$$

jednadžbu $x^3 + 2x^2 - 5x - 6 = 0$.

Rješenje: $r(A) = 2 \rightarrow$ zadovoljava!